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# Machine-Learning-Driven Digital Twin for Lifecycle Management of Complex Equipment

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**ABSTRACT** The full life cycle management of complex equipment is considered fundamental to the intelligent transformation and upgrading of the modern manufacturing industry. Digital twin technology and machine learning have been emerging technologies in recent years. The application of these two technologies in the full life cycle management of complex equipment can make each stage of the life cycle more responsive, predictable, and adaptable. This paper first proposes a technical system that embeds machine learning modules into digital twins. Next, on this basis, a full life cycle digital twin for complex equipment is constructed, and joint application of sub-models and machine learning is explored. Then, the application of a combination of the digital twin in maintenance with machine learning in predictive maintenance of diesel locomotives is presented. The effectiveness of the proposed management method is verified by experiments. The abnormal axle temperature can be alarmed about one week in advance. Lastly, possible application advantages of the combination of digital twin and machine learning in addressing future research direction in this field are introduced.

**INDEX TERMS** Complex equipment, digital twin, machine learning, operation and maintenance health management

## I. INTRODUCTION

The manufacturing industry is an important part of the global economy. With the development and integration of Big-Data, Internet of Things, Cloud Computing, Artificial Intelligence, and other advanced technologies into the manufacturing industry [1]–[5], the modern manufacturing industry has been rapidly developing. Intelligent manufacturing with core cyber-physical integration has become the weathervane of future manufacturing development [6], [7].

The essence of intelligent manufacturing is to perceive, collect, and analyze various data in the production process using the combination of advanced manufacturing technology and information communication technology to realize information depth perception, adaptive control, and self-optimization decision-making. Intelligent manufacturing has been implemented into the whole life cycle of manufacturing activities such as design, production, operation and maintenance management, recovery, and scrap. The emergence of

digital twin technology has provided a solution to connect the physical world with the information world while realizing data interaction and integration. As an important artificial intelligence type, machine learning can mine useful knowledge in data, providing decision support to users for solving problems by applying deep learning and processing to fusion information. In the recent development of intelligent manufacturing, Industry 4.0, the theories of digital twin and machine learning have been improved and widely used. Uzun *et al.* [8] employed a deep learning algorithm to establish the fuel consumption model of a jet aircraft and optimized the flight performance of the aircraft using the digital twin. Tao *et al.* [9] proposed a concept of the Digital Twin Shop-floor (DTS) to support intelligent manufacturing. The DTS includes four key components: Physical Shop-floor (PS), Virtual Shop-floor (VS), Shop-floor Service System (SSS), and Shop-floor Digital Twin Data (SDTD). Zhang *et al.* [10] developed a method combining digital twin and

stacked autoencoder (SAE) to monitor the product quality. Fang *et al.* [11] introduced a digital-twin-based job shop scheduling method; they used an innovative scheduling resource parameter updating method and a dynamic interactive scheduling strategy to achieve accurate real-time scheduling. Liu *et al.* [12] proposed a process evaluation method based on digital twin, which considers dynamic changes in processing conditions and uncertainty of available manufacturing resources. This method has been effectively implemented in key components processing of a marine diesel engine.

Complex equipment, such as an aerospace vehicle, complex electromechanical equipment, and railway transportation equipment, is the important basic equipment in the manufacturing industry chain. Product life cycle management and service of complex equipment are important for the transformation and upgrading of the global manufacturing industry [13]. The main objective is to establish a digital twin for complex equipment and manage the full life cycle of complex equipment by using a calculation model driven by both models and data. In this way, the current status of complex equipment could be better understood, monitored, tracked, and mastered. Therefore, it is necessary to provide required services based on information interaction feedback.

The main contributions in this study can be summarized as follows:

- Machine learning-based methods are used to improve the cognition and decision-making capabilities of a digital twin, and a full-lifecycle technology architecture, in which the computing model is embedded in the twin, is proposed.
- A suitable digital twin architecture for complex equipment life cycle management is introduced, and functions of the proposed machine learning-based method in the digital twin of complex equipment in each of the stages are discussed.
- The specific application of the proposed architecture is demonstrated in operation and maintenance health management case. Based on the series speed-axle temperature prediction model, the digital twin is used in the maintenance to analyze the reliability of the bearing state, and active maintenance of the CDD5B1 diesel locomotive is achieved.

The rest of this paper is structured as follows. Section II reviews related literature. Section III presents a digital twin technology system based on machine learning for complex equipment. Section IV proposes a whole-life-cycle digital twin architecture based on the above system. Section V presents a specific application case of the proposed method and framework in equipment health management. Section VI concludes the paper.

## II. LITERATURE REVIEW

Digital twin uses digital technology to describe and model physical entities in the real world. It has been used to simulate or dynamically present various characteristics, behavior, formation process, and performance of an object in different

time moments so as to achieve understanding, analysis, monitoring, and optimization of physical entities.

Machine learning represents data analysis technology that automates the construction of an analysis model. It can enable a system to learn and identify data patterns so as to obtain new knowledge or skills, reorganize the existing knowledge, improve performances, and make decisions under minimal manual intervention. Machine learning automates the analysis model construction.

In the industrial field, a digital twin has been considered the main tool for realizing intelligent manufacturing. Since the self attributes of machine learning can improve the performance of a digital twin, their integration could produce value-added capabilities for fault diagnosis, robot-assisted assembly, quality monitoring, job shop scheduling, and detailed management. At present, the combination of digital twins and machine learning has been used in many modern approaches and applications, as shown in Table 1.

The literature review explains several shortcomings and limitations of the existing studies. First, there have been fewer studies on machine learning application to the whole life cycle management of complex equipment within the scope of the digital twin, and there has been a lack of targeted digital twin management architecture for complex equipment. Second, the existing literature on complex equipment has been mostly focused on a certain stage, such as design, manufacturing, or operation and a particular use case. Therefore, there has been a lack of management theory and technical system covering the whole life cycle of complex equipment, resulting in disconnection of links, such as complex equipment design, processing and manufacturing, operation and maintenance, low reusability of design information, inability of real-time feedback of equipment operation status, and low efficiency of cycle iteration. Thus, it has been challenging to achieve a closed-loop. Third, there has been less discussion on methods integrating digital twin and machine learning, as well as the architecture design of adopting emerging technologies.

This paper aims to address the existing research limitations by proposing a complex equipment life cycle digital twin architecture based on machine learning. The proposed architecture establishes a digital twin model of the whole life cycle, uses data obtained from the physical entity, and employs the machine learning model to create a virtual model. After the generation of digital twins, data of physical and virtual twins are provided to the other machine learning models to achieve a given industrial goal, such as design optimization, manufacturing process dynamic planning, or PHM. Complex equipment management has requirements for full-cycle realization, multi-party coordination, and short cycle combination. To satisfy these requirements and achieve distributed processing and converged management, this paper innovatively proposes an integration architecture of digital twin, machine learning, and edge-cloud computing.

**TABLE 1. Presentation of combined application of digital twin and machine learning.**

| Ref.       | Approach                                                    |                                                                                                   | Application                                               | DT use-case /Implementation environment |
|------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------|
|            | AI-ML                                                       | Other features                                                                                    |                                                           |                                         |
| [10]       | Artificial neural network (ANN), stacked auto-encoder (SAE) | /                                                                                                 | Product quality                                           | CNC bending machine                     |
| [8]        | Deep learning                                               | /                                                                                                 | Fuel burn estimation, model performance updating          | Jet aircraft                            |
| [28]       | Deep learning                                               | Cloud digital twin and edge digital twin                                                          | Collaborative data management, data analytics in metal AM | Project MANUELA                         |
| [29]       | Feed-forward neural network (FFNN), Genetic Algorithm (GA)  | /                                                                                                 | Performance optimization                                  | Dew point cooler                        |
| [30]       | Process knowledge reuse network (PKR-Net)                   | /                                                                                                 | Intelligent process planning                              | Impeller                                |
| [31]       | Reinforce Learning                                          | /                                                                                                 | Geometric optimization                                    | Centrifugal impeller                    |
| [32]       | Deep transfer learning                                      | /                                                                                                 | Fault diagnosis                                           | Car body-side production line           |
| [33]       | Ant colony optimization                                     | /                                                                                                 | Robot optimization (avoiding obstacles)                   | Robots                                  |
| [34]       | Deep learning                                               | /                                                                                                 | Remaining useful life prediction                          | Casting roller                          |
| This paper | Machine learning                                            | Three-layer architecture, including the data layer, edge-cloud computing layer, and service layer | Lifecycle management (e.g., health management)            | Complex equipment (e.g., locomotive)    |

### III. PROPOSED DIGITAL TWIN TECHNOLOGY ARCHITECTURE

The establishment and application of the digital twin depend on the development and innovation of many technologies, such as perception control technology, data acquisition and transmission technology, massive data management and processing technology, and high-performance computing technology. The machine learning methods can be embedded into a digital twin technology system to establish two-way information connection and integration, where a two-way connection indicates a two-way relationship between the physical and virtual worlds and the digital twin of sub-models in the virtual space. This is helpful for the landing use and innovative applications of a digital twin. This paper presents a machine learning-based digital twin technology system for complex equipment. The proposed system is shown in Figure 1, where it can be seen that it consists of three layers, namely, data layer, edge-cloud computing layer, and service layer. It should be noted that there is a progressive relationship between the three layers. Starting from the edge-cloud computing layer, the construction of each layer is based on the previous layers, providing further enrichment and expansion of the previous layers' functions. The functions of the three layers are explained in detail in the following.

#### A. DATA LAYER

The data layer represents the basis and underlying support of the entire digital twin technology system. It includes the data acquisition module and the data transmission module. The data needed for the operation of the whole digital twin system are provided by the data acquisition module, and these

data include real-time state sensing data, historical operating data, and user interaction data. Data collection equipment (e.g., RFID tags and sensors) collect data from the physical site. High-performance sensors and distributed sensing technology ensure the accurate acquisition of real-time physical characteristics of the equipment. The historical operation data of a system include historical information on the design, manufacturing, service, and recycling of the same type systems. All historical information is filed and stored for subsequent usage. The user demand interaction data are obtained through users' interaction with physical products via active behaviors characterized by certain setting preference parameters and feedback on usage. These data reflect user experience and suggestions. All these data are transmitted to the upper system via the data transmission module to support the operation of the upper system.

The application of 5G and other advanced technologies shortens data transmission time and improves the communication service quality and real-time follow-up performance of a digital twin system. For example, the 5G communication technology has the advantages of high bandwidth, low delay, strong connection, and high reliability [14].

#### B. EDGE-CLOUD COMPUTING LAYER

The realization of the assistant reality function of a digital twin depends on a powerful computing function. The computing layer is constructed based on edge cloud collaboration, and it is composed of the modeling algorithm module and the computing reasoning module. After the modeling algorithm module obtains data provided by the data layer, it uses a data-driven method to establish a structured knowledge network data model with full links, multiple disciplines,

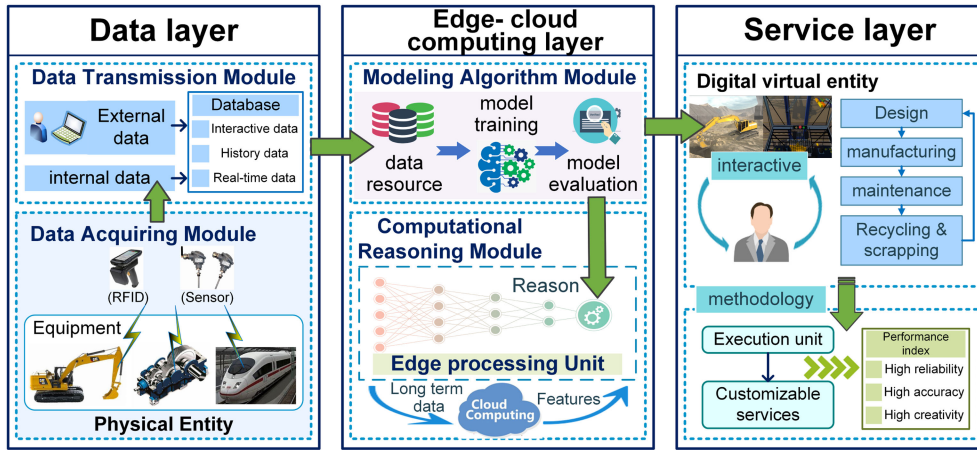


FIGURE 1. The ML-based digital twin technology system of complex equipment.

and cross-domains. This layer combines intelligent learning and optimization algorithms such as machine learning-based algorithms to establish data mining and knowledge discovery models. The computing reasoning module uses a super-realistic data representation model to perform knowledge learning and reasoning and provides the multi-angle decision-making results and analysis conclusions needed for the realization of the functional layer of the digital twin.

The real-time, short-period data analysis task is submerged in complex equipment, which provides better support for intelligent processing and execution of local business. The non-real-time, long-period data, and effective data obtained from edge computing are uploaded to the cloud platform for further processing and feature extraction. The database is expanded and optimized. This approach ensures advantages in areas such as periodic maintenance, overall decision-making, and feedback-based optimization of virtual "small cycles" production and actual "large cycles" production.

### C. SERVICE LAYER

The service layer represents a customizable service window that can provide functional services for variable requirements of the full life cycle management of complex equipment. Considering different optimization perspectives, the functional layer provides services directly to reality based on the reasoning results of the computing layer. The service meets requirements for multiple performance indicators, including high reliability, high accuracy, high real-time, and high-value creativity. This is the embodiment of the core value of the digital twin system.

Complex equipment has the characteristics of personalization and customization, and its digital twin serves manufacturers and users. The service layer is based on the massive, multi-dimensional data of the data layer and the powerful information interface of the computing layer. It can customize value-added business functions according to the actual needs and can also improve the intelligence level and user experience of manufacturing enterprises.

## IV. DIGITAL TWIN ARCHITECTURE FOR COMPLEX EQUIPMENT

One of the primary objectives of a digital twin is to integrate cyberspace and physical space [15], which is to apply the digital twin to the life cycle management of complex equipment and understand, analyze, monitor, and optimize the state or behavior of physical equipment. This can be realized through virtual interactivity and feedback, data mining and analysis, and multi-source data association and fusion [16], [17], and this approach can satisfy the requirements of complex modern equipment manufacturing. A product digital twin of complex equipment is constructed to perform different functions, including geometric perception, behavior simulation, and process recovery of physical equipment and its architecture, as shown in Figure 2. The proposed architecture establishes a full life cycle data center for complex equipment and uses a single data source to realize the bidirectional connection between physical and information spaces and data communication between all stages of the full life cycle. The digital twin sub-models of design, manufacturing, maintenance, recycling, and scrapping are integrated into the digital twin of complex equipment. These models provide necessary model support for the life cycle management of complex equipment. These four sub-models complement each other and evolve together; they use the analytic hierarchy process (AHP), Bayesian estimation, expert system, D-S reasoning method, artificial intelligence, and other related methods to analyze and utilize data sources. This provides the high-potential added value introduced by the virtual data and services behind physical products and realizes dynamic fusion for updating and monitoring. Thus, it is essential to perform the whole life cycle process in real-time through intelligent development [35].

### A. DIGITAL TWIN IN DESIGN

Product design is based on the target function of the design object and user requirements. After discussion, analysis, and design processing, the requirements are transformed into a specific text or graphic expression to provide reasonable



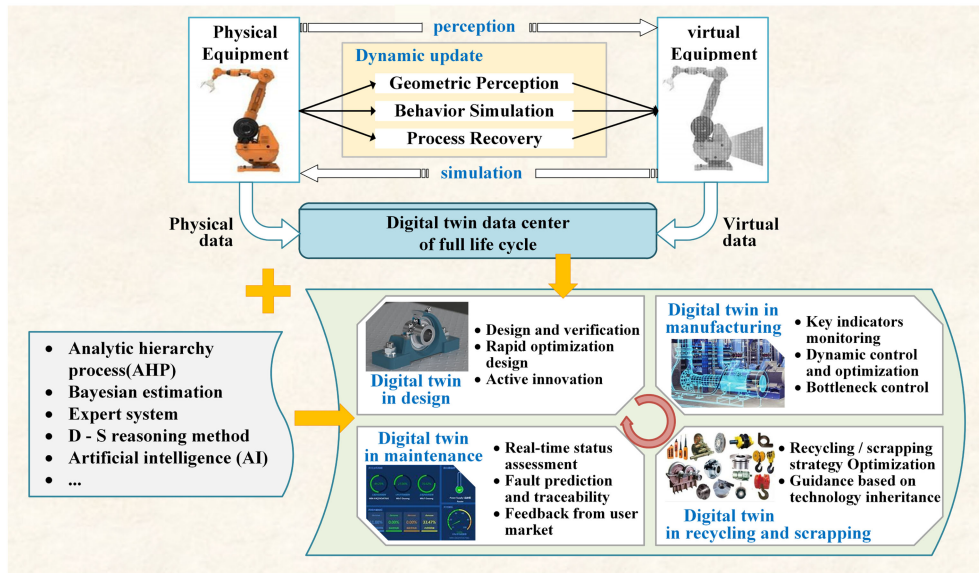


FIGURE 2. Digital twin architecture of complex equipment.

planning for the production stage [18]. Driven by the product's digital twin data, the Digital Twin in Design (DTD) provides a two-way interactive connection between physical and virtual products for data sharing. It also strengthens the synergy between these two and continuously explore innovative, unique, and valuable product optimization design schemes. Thus, it can achieve dynamic innovation that meets the customized requirements ahead of the time.

The digital twin is not only a digital product display model; it can improve the accuracy of the design and can assign the behavior of the actual product to the corresponding virtual product. In addition, it can verify the performance of a product in a real environment through simulation. The constructed design originates from practice and serves practice. The DTD emphasizes the virtual-physical integration of the complete life cycle. It establishes links between products, production tools, environments, and processes and integrates information of different stages of the product life cycle into the design phase while providing a coherent overview of the development cycle. Therefore, the DTD can promote decision-making and improve design quality and efficiency in many aspects through its various functions, thus, improving and optimizing products and their manufacturing systems.

## B. DIGITAL TWIN IN MANUFACTURING

It is necessary to construct the Digital Twin in Manufacturing (DTMF) before performing actual production tasks. It is also necessary to extract manufacturing-related process information from the DTD along with associated manufacturing resources information, such as information on processing equipment, process parameters, tooling, and moulds. To realize the DTMF, it is important to simulate the production process of complex equipment under agreed conditions through virtual production in advance. This process can predict

bottlenecks of resources, workpieces, and stages. Thus, in this way, bottleneck control can be achieved, and the speed and accuracy of a new product can be improved during the production process through the redistribution of bottleneck resources and optimization of component combination and operation sequence performed before production.

The customized and sophisticated manufacturing process of complex equipment requires effective process planning and key indicator monitoring [19]. In the manufacturing stage, the connection is established between the manufacturing entity and the virtual product by the DTMF. The measured data related to the product production, processing, and quality control processes, such as geometric measurement, vibration measurement, force measurement, and progress data, are mapped to the virtual product and displayed in real-time. This process ensures a comprehensive collection of production factors and manufacturing information and achieves online process control as well as monitoring based on the actual production data.

The manufacturing process is characterized by dynamics and uncertainties, such as tool wear, aging of processing equipment, and changes in material properties, which are caused by environmental conditions. Despite careful planning, these situations can cause a gap between the actual production results and expected results [20]. The DTMF helps to integrate process planning data with the measured production data while monitoring key manufacturing parameters. Therefore, when an abnormal situation occurs, such as a violation of plans or discovering a better production plan through learning, the intelligent decision-making module provides the corresponding treatment and scheme adjustment timely. The new adjustment scheme is fed back to the manufacturing entity to improve the manufacturing quality and to realize dynamic control and self-optimization, thus achieving the purpose of the virtual control.

### C. DIGITAL TWIN IN MAINTENANCE

Complex equipment typically has the characteristic of high cost, which is related to costly maintenance. However, both users and enterprises expect to achieve the maximum value of the equipment at the lowest possible cost. In the use and service phase of a product, the Digital Twin in Maintenance (DTMT) is used to track and monitor the product status in real-time. The DTMT acquires real-time data that characterize the quality and functional status of the monitored equipment during operation, such as engine rotor speed, vibration value, lubricating oil temperature, and exhaust pressure, by using smart sensors. It also helps in constructing a remote monitoring system with quantitative indicators and establishes a hierarchical health management system for complex equipment parts, subsystems, systems based on equipment historical use and maintenance data. It analyzes and predicts equipment operating conditions, remaining life, and potential failures and assets failure time in advance. It also performs beforehand maintenance to improve the efficiency of spare parts management and to reduce or avoid customer losses and problems caused by unexpected equipment shutdown. A virtual product can adjust itself to adapt constantly to changes in the physical space conditions. It reflects and evaluates the health status of physical service products through virtual products. In this way, the risk of manual evaluation using a one-way physical information flow is reduced. The insights obtained by the application of the DTMT can also be used to improve product design and manufacturing processes.

For physical equipment with a fault or a quality problem, it is better to use a virtual product to quickly locate the fault or the problem. This approach can also be combined with the historical operation data before the failure and can both provide the simulation traceability of a virtual product and analyze the causes of quality problems from their root. It is also evident to develop discovery methods and solutions to the problem of the same type and to verify the feasibility of these solutions. It also feeds the final results back to the physical space to achieve the early warning and troubleshooting.

The application of complex equipment has a strong driving force for both enterprise production and social progress. The rationality of complex equipment design parameters and operating parameter settings, as well as the adaptability to different working conditions, determine the functional level, quality advantages, and customer satisfaction of equipment. Equipment providers collect real-time data during product service by the DTMT. Using these data, they construct an empirical model for parameter optimization for different application scenarios and provide customers with guidance on its configuration. This can improve user satisfaction, enhance user experience, and maximize the functionality of the equipment. At the same time, customers can connect with the product through the DTMT and thus provide positive feedback to manufacturers. By considering the customer preferences, manufacturers can obtain insight into potential, future, real-time demands of customers for products and thus can meet them in advance. Therefore, it can avoid making

the research and development decisions from offsetting market demand, shorten the design time and introduction cycle of new products, and enhance market competitiveness.

### D. DIGITAL TWIN IN RECYCLING AND SCRAPPING

The application of the Digital Twin in Recycling and Scrapping (DTRS) meets the requirements of green ecology and sustainable development [21]. The *out-of-service* equipment record data related to scrapping and recycling, such as reasons for scrapping and recycling, expected product life, and actual life, can be obtained through digital twin. The DTRS provides a reference for product scrapping and recycling operations by relative data analysis. The related data include material data of parts (e.g., scarce resources, recyclable materials, or toxic materials) in the DTD and information related to the working environment of components and equipment (e.g., working conditions such as exposure to uranium and other radioactive elements) in the DTMT. When the physical product is scrapped or recycled, the corresponding virtual product is kept and archived in the digital form. Following the concept of technological inheritance, the production and application data of the previous generation are passed to the next generation. Historical information from different stages of the product life cycle is integrated into the design stage, forming the closed-loop management of the product life cycle data. Based on specific product requirements, it guides the design improvement and functional innovation of the next generation and increases the efficiency of new product development and production. The historical normal-operation and failure data of scrapping and recycling equipment can provide a helpful basis for quality analysis and health management of the same or similar product groups.

## V. CASE ANALYSIS

This section presents the application example of the proposed complex equipment management method based on machine learning and DTMT, to verify the effectiveness of the proposed method.

### A. DTMT APPLICATION TO OPERATION AND MAINTENANCE OF HEALTH MANAGEMENT

Complex equipment represents important equipment support for modern economic industries. However, unexpected shutdowns caused by failures can affect both the economy and society. Therefore, the health management of complex equipment during its service is particularly important. Using a combination of the DTMT and deep learning technology, this study performs real-time monitoring and predictive evaluation of equipment status to improve stability and reliability of the equipment operation [22], [23]. As shown in Figure 3, mapping of the physical operating status data of the equipment entity to the DTMT is performed by the synchronous simulation operation in the virtual space. This enables real-time monitoring of equipment status and simulation of future operation and avoids interference of redundant monitoring data on equipment operation status. Data denote the basic

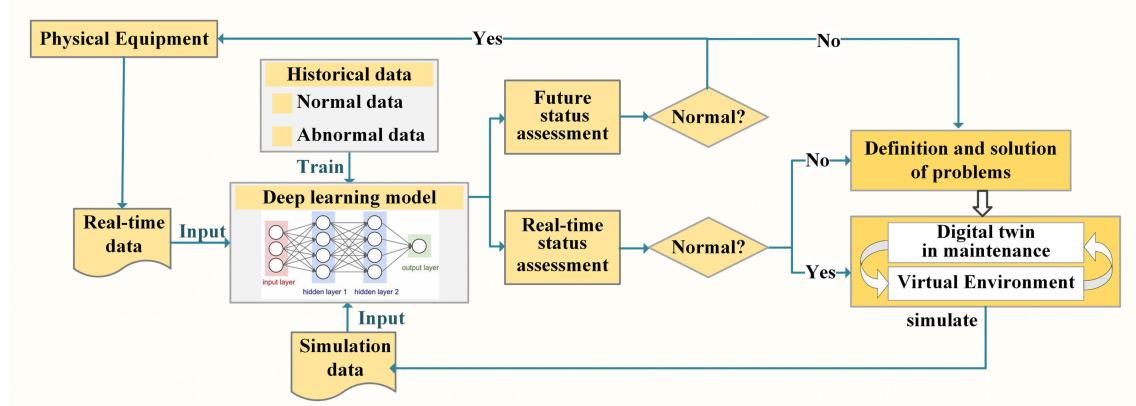


FIGURE 3. Application of the DTMT to the operation and maintenance health management.

and core driving force for the DTMT application to health monitoring and fault prediction of real equipment. A multi-layer nonlinear deep learning-based network structure has advantages in discovering useful information and essential characteristics of data and has high accuracy and completion degree in tasks like fault identification and prediction. The hierarchical abstraction capabilities of data features of deep learning algorithms can be useful to extract differences in high-dimensional association features between the monitoring data under normal and abnormal conditions using the historical data. It establishes association rule models between data and failures to monitor and predict the behavior and status of equipment. When an abnormality in equipment operation is detected by the real-time data analysis, i.e., when a fault has occurred, it automatically identifies and classifies the fault based on the correlation model. It analyzes the type and cause of the detected fault, formulates a maintenance plan fast, and restores or guarantees the normal operation of the monitored equipment. In case of simulation data anomalies at the time of a potential fault in the DTMT, it inputs the corresponding solution to the DTMT for simulation validation. If according to the simulation results, the monitoring data return a normal value, that indicates that the potential fault has been eliminated. Then, the corresponding decision-making plan is sent to the physical entity to guide the maintenance process, and prior repair is realized. This can reduce the equipment maintenance cost and uncertainty in the operation process and can achieve self-perception, self-prediction, and self-decision of equipment.

### B. HEALTH MANAGEMENT MECHANISM OF DTMT

Monitoring and changing trend prediction of characteristic parameters of main components help to optimize the complex equipment control. In this way, the operation state of complex equipment can be predicted better. Bearing, as a key component of transmission, exists in the gearbox, axle box, motor, and other key parts of complex equipment; it plays an irreplaceable role in maintaining motion accuracy and improving mechanical efficiency. Therefore, fault diagnosis and prediction of bearings could significantly reduce the maintenance

cost and time. Meanwhile, bearing temperature is an effective index that can directly reflect the bearing condition, and the temperature characteristics have the advantages of thermal inertia and strong anti-interference ability. Therefore, this study selects the temperature characteristic of bearing to judge whether a fault has occurred or not and then performs the health assessment. A locomotive is the main power provider in the railway transportation system, and it is an important piece of equipment that affects traffic safety. In this study, the running gear of the CDD5B1 diesel locomotive is taken as a research object, and the DTMT is used for predictive maintenance of diesel locomotive bearings. The specific implementation process is presented in Figure 4, where it can be seen that the process consists of three main stages: real-time data collection, establishment of a calculation model based on the DTMT, and bearing health assessment.

#### 1) DATA ACQUISITION AND DATABASE

The accuracy of fault prediction and health management depends on the relevance, adequacy, and quality of the data used. Therefore, it is necessary to establish a predictive maintenance database for the DTMT using different data, such as historical data from the same source, historical maintenance data, and real-time operating data. The characteristics of the train operation data are periodic during the operation cycle. After being denoised and normalized, the train operation data of the same type are uploaded to the cloud and stored as sample data for future use. Homologous historical data are backed up and updated on the cloud in time to ensure data consistency. Sink data are sent to the edge of the digital twin and used when needed. Historical maintenance data include detailed information on the maintenance activities performed by equipment, replaced components, and failures. According to the requirements for dynamic health assessment of the locomotive running department, this study establishes a sensor network system to percept and capture real-time data.

#### 2) PREDICTION MODEL BASED ON DTMT

The overall framework of the prediction model based on the DTMT is presented in Figure 5, where it can be seen that it



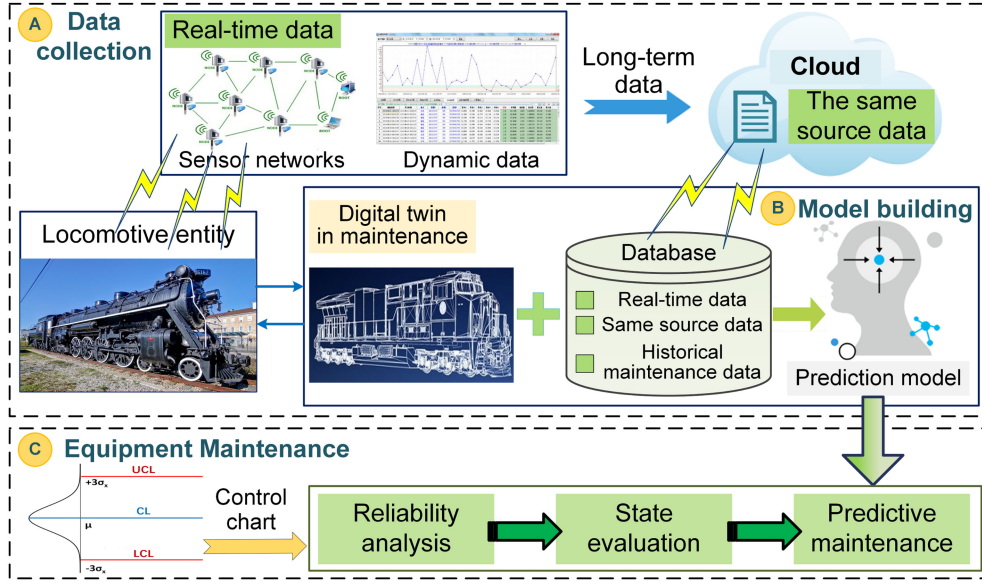


FIGURE 4. Health management implementation process based on the DTMT.

consists of three parts: the data preprocessing part, the single model prediction part, and the model integration part.

**Part A.** Preprocessing of the original data. Identify and delete duplicate values and outliers in the original data; use the Lagrangian interpolation to fill the missing values; use the *Savitzky-Golay* filter to denoise and normalize the data. The original data are divided into training and test sets. The training set is used to train the proposed model, and the test set is used to test the performance of the trained model.

**Part B.** Single model prediction results. Use three machine learning methods, namely, the *LASSO* [24], *SVR* [25], and *XGBoost* [26], to design an axle temperature prediction model in the DTMT and obtain the prediction results of the sub-series.

**Part C.** The combination prediction results. Obtain the final prediction result by integrating the prediction results of the three machine learning networks. In this study, the model set is realized by setting the weight coefficients of the prediction results of the learning networks. The corresponding calculation formulas are as follows:

$$\hat{T}(m) = w_1 \hat{T}_1(m) + w_2 \hat{T}_2(m) + w_3 \hat{T}_3(m), \quad (1)$$

where  $w_i (i = 1, 2, 3)$  denotes the weight coefficient of a learning network  $i$ ,  $\hat{T}_i(m) (i = 1, 2, 3)$  denotes the prediction

results of a learning network, and  $\hat{T}(m)$  represents the sensor data output by the DTMT.

### 3) BEARING HEALTH STATUS ASSESSMENT

The normal axis temperature data conform to the normal distribution. The nature of the normal distribution shows that the probability of data appearing outside the three standard deviations of the mean is only 0.2%, which is very small. According to the principle of probability theory [27] "taking small-probability events as practically impossible," it can be known that the occurrence of such an event is due to abnormal causes that make the overall distribution deviate from the normal position, indicating that a failure has occurred. Therefore, the *X-S* control chart can be used to control the normal temperature range of the axle with the upper control limit of  $\mu + 3\sigma_x$  and the lower control limit of  $\mu - 3\sigma_x$  (the two red lines in Figures 18, 20 and 22 are control lines).

## C. OPERATION AND HEALTH MANAGEMENT APPLICATION

The on-board safety protection system loaded on the CDD5B1 diesel locomotive collects parameters that can reflect the operating status of the monitored target in real-time and provides long-distance transmission and anti-interference protection. The monitoring status information includes but is not limited to the vehicle number, axle number, and temperature information, alarm prompt information, and vehicle speed. In addition, there are six axles in the locomotive, each of which has six temperature measurement points. When the temperature of a certain axis reaches or exceeds 90 °C, the system will issue an alarm. On March 26, 2018, during the operation of the dual-locomotive traction train of CDD5B1-5109 +5111 locomotive, the axle temperature detection device of the 5109 locomotive announced 51 bearing-temperature-rise alarms, and the temperature was 89 °C. After the on-site

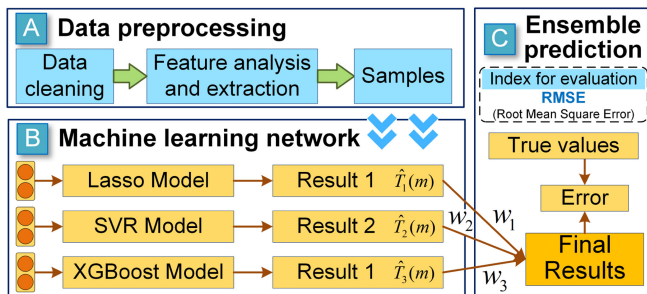


FIGURE 5. The flowchart of the DTMT-based prediction model.

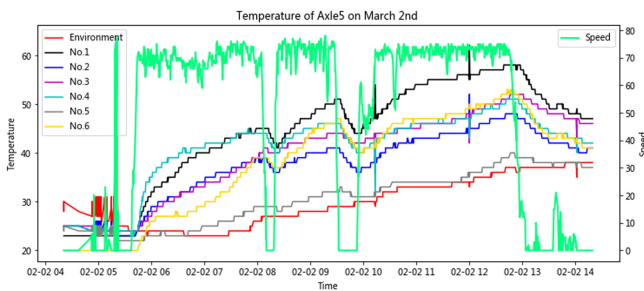


**TABLE 2. Characteristic terms of a single axle.**

| term number | term name                                           |
|-------------|-----------------------------------------------------|
| 1           | The environment temperature/°C                      |
| 2           | Main generator temperature/°C                       |
| 3           | GPS speed/km/h                                      |
| 4-9         | Temperature of measuring points 1-6 of axis $X$ /°C |

**TABLE 3. Characteristic terms of a single measuring point.**

| term number | term name                                          |
|-------------|----------------------------------------------------|
| 1           | The environment temperature/°C                     |
| 2           | Main generator temperature/°C                      |
| 3           | GPS speed/km/h                                     |
| 4           | Temperature of measuring point $M$ on axis $X$ /°C |

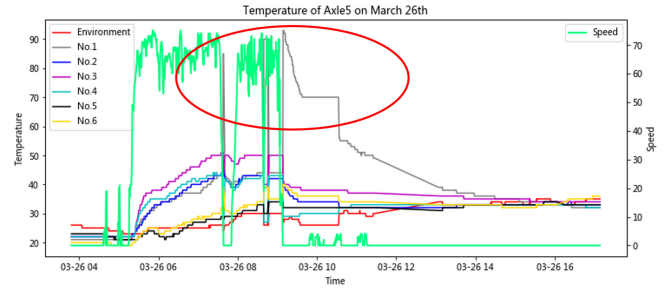
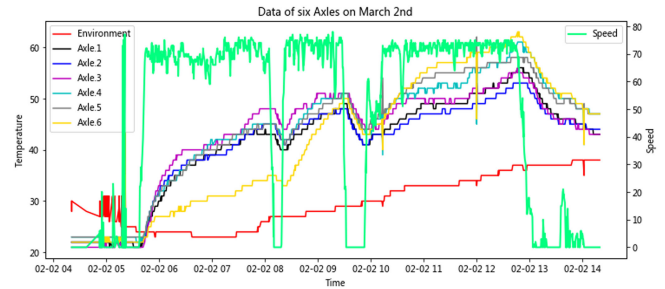
**FIGURE 6. Normal data of the No. 5 axle over time.**

inspection by maintenance personnel, it was found that the outer bearing cylinder of the axle box had felt into the axle box cover, and the cage was severely deformed.

The axle temperature alarm device can announce an alarm only when abnormal values of physical quantity have been detected to remind drivers to stop and check the vehicle, which causes a postponement. The bearing faults denote cumulative processes, and in this study, the predictive maintenance method based on the DTMT is used to perform the health management of the bearings of the CDD5B1 diesel locomotive and to pre-judge faults in advance.

In this paper, the data used for case implementation is the original data in Excel format provided by CRRC Qishuyan Locomotive Co., LTD. (Changzhou, Jiangsu, China). The original data includes the normal (no alarm of axle temperature detection device) data of CDD5B1-5111 locomotive in March 2018 and the fault (alarm of axle temperature detection device) monitoring data of CDD5B1-5109 locomotive before March 26, 2018. The axle temperature monitoring and alarm system of a diesel locomotive has 67 data monitoring items. Data is collected about every 30 seconds, but repeated collection may occur at the same time point. According to the daily running time of locomotive, 1,000-2,000 data items may be collected, with a total of nearly 10,000 data points.

The CDD5B1-5111 locomotive runs synchronously with the CDD5B1-5109 locomotive, and their two states are similar. Therefore, the historical operation data of the CDD5B1-

**FIGURE 7. Abnormal data of the No. 5 axle over time.****FIGURE 8. Normal data of the No. 1 point of six axles over time.**

5111 locomotive under normal conditions can be used as homologous historical data. These data are added to the CDD5B1-5109 locomotive DTMT database to build the axle temperature prediction model.

When a single axle is taken as the monitoring object, the feature items that can be selected are listed in Table 2.

When a single measurement point is taken as the monitoring object, the feature items that can be selected are listed in Table 3.

For a single axle as a monitoring object, the changes in the normal data of the No. 5 axle of the CDD5B1-5111 locomotive and the abnormal data of the 5,109 locomotive over time are presented in Figures 6 and 7, respectively. For a single measuring point as a monitoring object, the changes in the normal data of the No. 5 axle of the CDD5B1-5111 locomotive and the abnormal data of the 5,109 locomotive over time are presented in Figures 8 and 9, respectively.

In Figures 6 and 8, it can be seen that under normal circumstances, the changing trends of the temperature of the same measuring points on the same axle and the same measuring point on different axles are roughly the same. Also, there is a certain correlation with the environmental temperature. When a fault occurs, the temperature data of a certain measuring point of the faulty axle deviates from the temperature change range of the remaining measuring points of the axle. Namely, it rises sharply, resulting in an "outlier" phenomenon. In this study, a temperature-speed model is established based on measured point 1 of axle 5, and the measured point temperature is used as an effective index to judge the working status of the No. 5 axle. In this way, the working conditions and operational safety of the entire diesel locomotive can be evaluated.

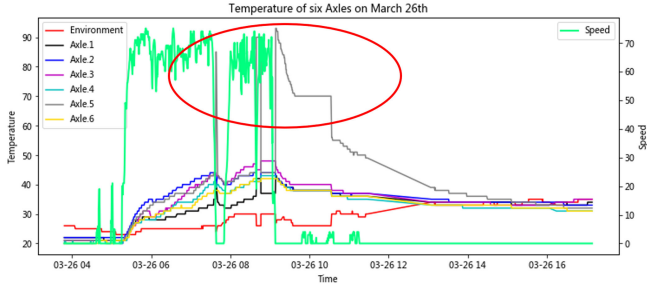


FIGURE 9. Abnormal data of the No.1 point of six axles over time.

Figure 10 shows the temperature change of measuring point 1 on the six axles with the ambient temperature and locomotive speed in the week before the fault has occurred; it can be seen that although there was no alarm from the axle temperature monitoring device before March 26, the data of the No. 5 axle 1 measurement point had been in an "outlier" state for a long time. Thus, it can be inferred that the axle temperature at this time had deviated from the normal axle temperature data. Although the temperature did not exceed the normal temperature range and predefined maximum temperature, there might be a potential failure since at that time, the temperature could not correctly reflect the safe operation of the locomotive. These results provide feasible support for simulating the correct temperature and predicting potential failures using the DTMT.

The results show that there is a certain delay between the axle temperature change and the speed change. In this study, a time-series index is used to select non-parking phase data (i.e., GPS speed of zero) from the overall locomotive operation data. The environmental temperature and the main generator temperature denote the environmental data; the GPS speed is used as control data. The extracted data are used as input data of the axle temperature prediction model. Considering the delay between the axle temperature change and the speed change, the input variables are set to the current speed and the speed of the previous four-time steps. Aiming to reduce the impact of the delay, the DTMT considers the operating conditions of the locomotive, such as starting, accelerating, running smoothly, braking, or decelerating.

In this study, the monitoring data of the CDD5B1-5111 locomotive are randomly selected from the normal operation data of one day, and the axle temperature-speed prediction models are designed using the *LASOO*, *SVR*, *XGBoost* algorithms. The series method is employed to determine the model weight coefficients. The modeling process is shown in Figure 11. From the overall data, 10% of the samples are randomly selected as a test set, and the prediction results of different models on the test set are shown in Figures 12, 13, 14, and 15. All experiments are performed on the Python 3.8 platform installed on a personal computer.

The Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Coefficient of Deviation (R2) are used to evaluate the performance of the three single prediction models, the *LASOO*, *SVR* and *XGBoost* models, and their combined models. The RMSE, MAE and R2 is calculated by:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2} \quad (2)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |p_i - y_i| \quad (3)$$

$$R2 = \frac{\sum_{i=1}^n (p_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (4)$$

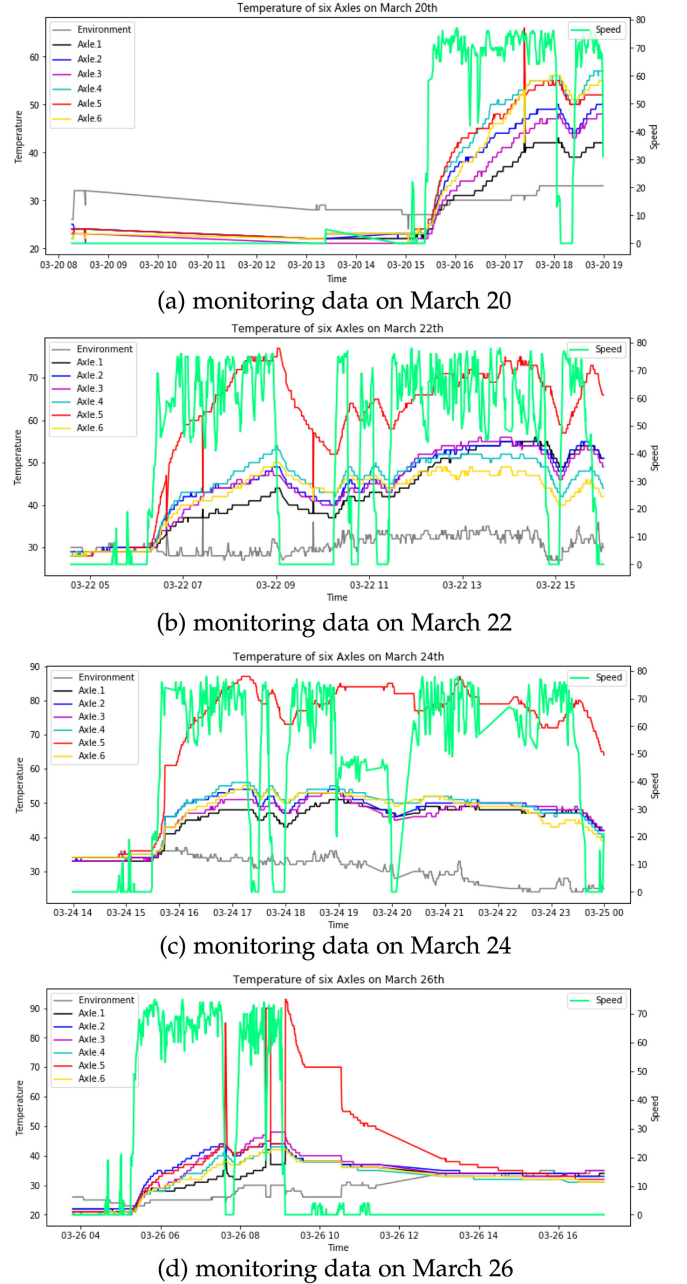


FIGURE 10. Monitoring data of the CDD5B1-5109 locomotive a week before the failure has occurred.

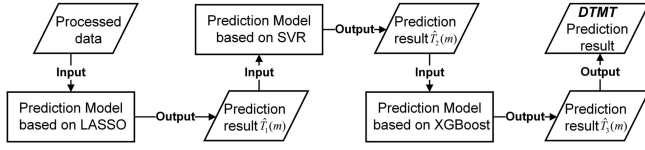


FIGURE 11. The flowchart of the series prediction model.

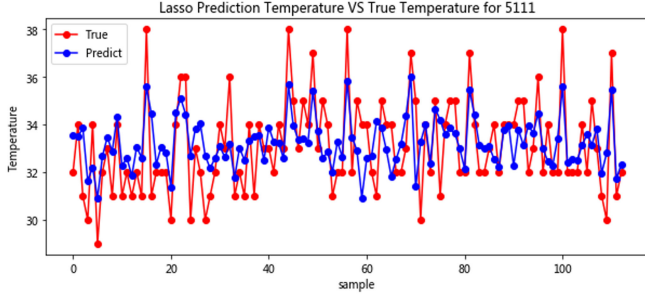


FIGURE 12. Prediction results of the *LASSO* model.

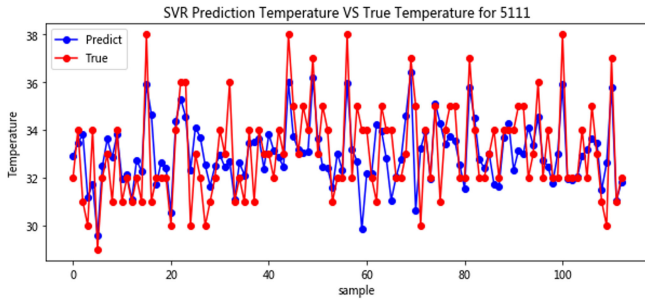


FIGURE 13. Prediction results of the *SVR* model.

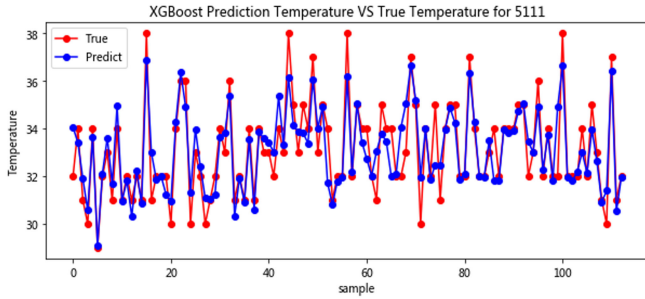


FIGURE 14. Prediction results of the *XGBoost* model.

where  $m$  denotes the number of predicted samples,  $\bar{y}$  is the mean of the true values of all the samples,  $p_i$  and  $y_i$  are the predicted value and true value of the  $i$ -th data sample.

The RMSE, MAE, R2 values of the four prediction models are given in Table 4.

As shown in Table 4, the RMSE and MAE of the series combination forecasting model is the smallest, and its R2 value is closest to 1. These results show that the forecasting accuracy of the series combination forecasting model is the highest among all the models. Therefore, a series

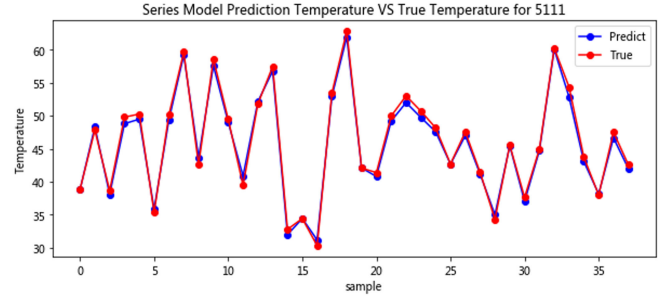


FIGURE 15. Prediction results of the proposed series model.

TABLE 4. The *RMSE* of the four prediction models.

|                                     | RMSE   | MAE    | R2     |
|-------------------------------------|--------|--------|--------|
| <i>LASSO</i> prediction model       | 1.5035 | 1.0597 | 0.6926 |
| <i>SVR</i> prediction model         | 1.5148 | 1.1361 | 0.6475 |
| <i>XGBoost</i> prediction model     | 0.9364 | 0.6815 | 0.7283 |
| Series Combination prediction model | 0.7105 | 0.4333 | 0.8739 |

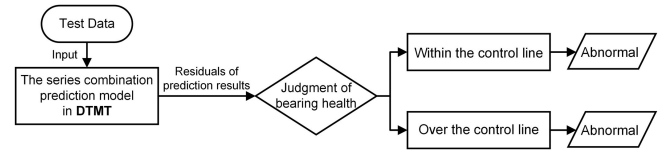


FIGURE 16. The block diagram of the verification process of the combined prediction model.

combination prediction model based on the DTMT is designed to predict a failure and warn about the axle temperature of a diesel locomotive.

The normal axle temperature data and fault data are used to verify the established axle temperature–vehicle speed series combination prediction model. The block diagram of the verification process is shown in Figure 16.

The prediction results and residual error distributions of the axle temperature during a normal operating day of the CDD5B1-5111 locomotive are shown in Figures 17 and 18, respectively. The prediction residuals of the normal data are all within the control range. This result shows that the interior of the locomotive system is in line with the normal speed–axle temperature mechanism. There is no undetected problem of failure, and the locomotive is safe and reliable.

The predicted results of the failure data and the distribution of residual error when the CDD5B1-5109 locomotive failed on March 26 are shown in Figures 19 and 20, respectively. The prediction residuals of the fault data have mostly exceeded the control line. This shows that at this time, the interior of the locomotive's system did not meet the normal speed–axle temperature mechanism, which resulted in a failure alert.

Through the prediction of the axle temperature under the normal and failure conditions of the locomotive, the correctness of the proposed model for the axle health status

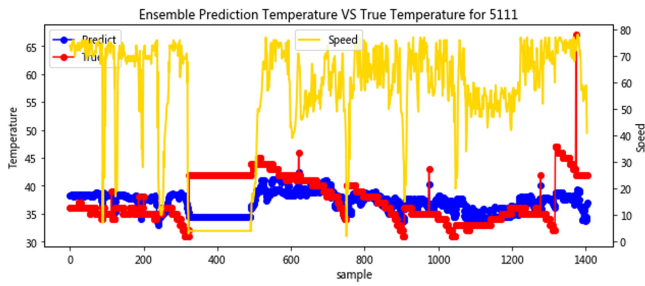


FIGURE 17. Normal data prediction results.

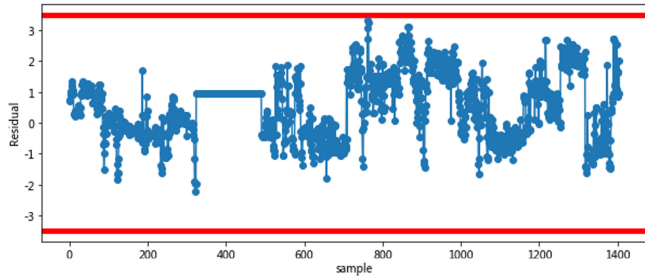


FIGURE 18. Normal data prediction result residuals.

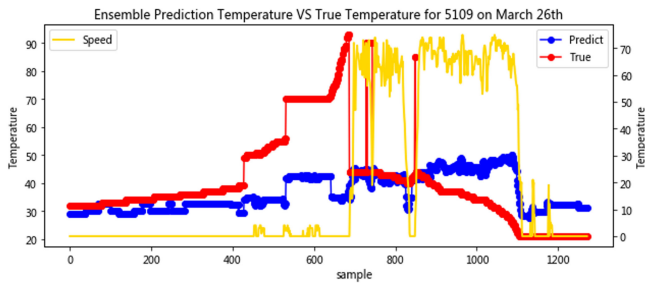


FIGURE 19. Abnormal data prediction results.

assessment has been verified. When the DTMT is used to predict the axle temperature of the CDD5B1-5109 locomotive on March 20, the prediction results and residual distribution are as shown in Figures 21 and 22, respectively.

As shown in Figures 21 and 22, although the axle temperature monitoring device did not alarm on March 20, the residual error of the axle temperature prediction result has partly exceeded the normal control range, which was abnormal. The first warning occurred during the first acceleration of the locomotive, after which frequent abnormal warnings occurred. This shows that the interior of the locomotive at that time has failed or has a failure trend, so it needed to be stopped for maintenance purposes.

Therefore, by using the management method based on a combination of digital twin and machine learning, the abnormal axle temperature can be alarmed about one week in advance, hidden problems can be discovered, and maintenance can be performed according to the situation beforehand. In this way, the occurrence of function failures can be reduced, and the safety of the locomotive operation can be ensured.

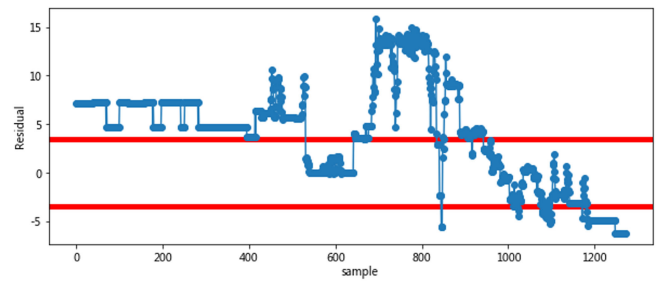


FIGURE 20. Abnormal data prediction result residuals.

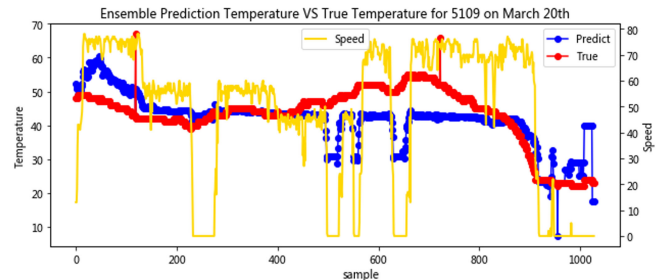


FIGURE 21. Axle temperature prediction results on March 20.

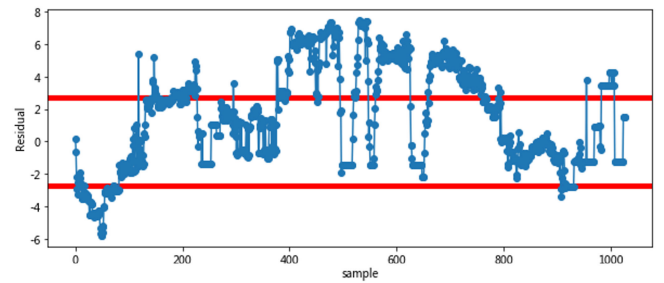


FIGURE 22. Axle temperature prediction results residual on March 20.

In summary, under the background of a digital twin, this paper proposes a machine learning-based management method of complex equipment. The application example demonstrates the effectiveness and practical value of the proposed method in health management maintenance. The proposed method can be extended to the field of life cycle management.

## VI. CONCLUSION

Digital twin technology is one of the emerging digitalization technologies, which is capable of transforming physical parameters or processes into their corresponding quasi-real-time digital copies that operate synchronously. It provides monitoring and simulation analysis in the virtual space. Digital twin provides a new solution to the cyber-physical integration with complex dynamic systems and introduces new opportunities and challenges to the lifecycle management and optimization of complex equipment. The combination of machine learning algorithms with digital twin technology can help to mine the rules and internal relationships of data and



discover the nature of the existing problems. Through the integration of data and knowledge at each stage, the collaboration and optimization of the whole life cycle of equipment can be achieved. The application of digital twin technology and machine learning to the field of smart manufacturing can help solve the problem of much data but insufficient information and disconnection between processing, manufacturing, operation, and maintenance operations. The proposed model has broadened the prospects for future industrial development. In this study, a full-lifecycle digital twin of complex equipment is designed based on the data dimension, and the digital twin sub-models are described in the phases of design, manufacturing, maintenance, scrapping, and recycling. Four application examples of deep reinforcement learning algorithms embedded with the digital twin models at each stage are constructed, improving the cognitive, inferential, and decision-making abilities of digital twins.

In this study, the architecture of embedding the machine learning-based computing module into the digital twin is introduced. Based on the data dimension, a full-lifecycle digital twin of complex equipment is constructed. The integration and application of the digital twin sub-models and machine learning in the stages of design, manufacturing, operation, and maintenance, and scraping or recycling, are analyzed. Then, the actual application case of the machine learning-based algorithm embedded into the DTMT to provide predictive maintenance services is studied. The verification results prove the feasibility and effectiveness of the proposed digital twin modeling method.

This study has aimed to explore and discuss the possibility and methods of combining machine learning with digital twin technology. At present, the research and application of digital twin technology have still been in the development stage, so further research is necessary. In the future, the digital twin model could be further refined at each stage, and multiple types of time-series data could be used to construct a digital twin for the entire life cycle of complex equipment. Further, virtual models could be used to calculate untested or difficult-to-test parameters, and more comprehensive monitoring of equipment status could be performed. Furthermore, more complete in-depth prediction and decision-making models could be designed to enhance data analysis and digital twin applications in Industry 4.0.

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